

R1-2 A Detailed Analysis of LLM Execution on IMAX3 and Initial Evaluation of IMAX4 Prototype for Server Environment

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Background & Objective

The LLM Energy Challenge

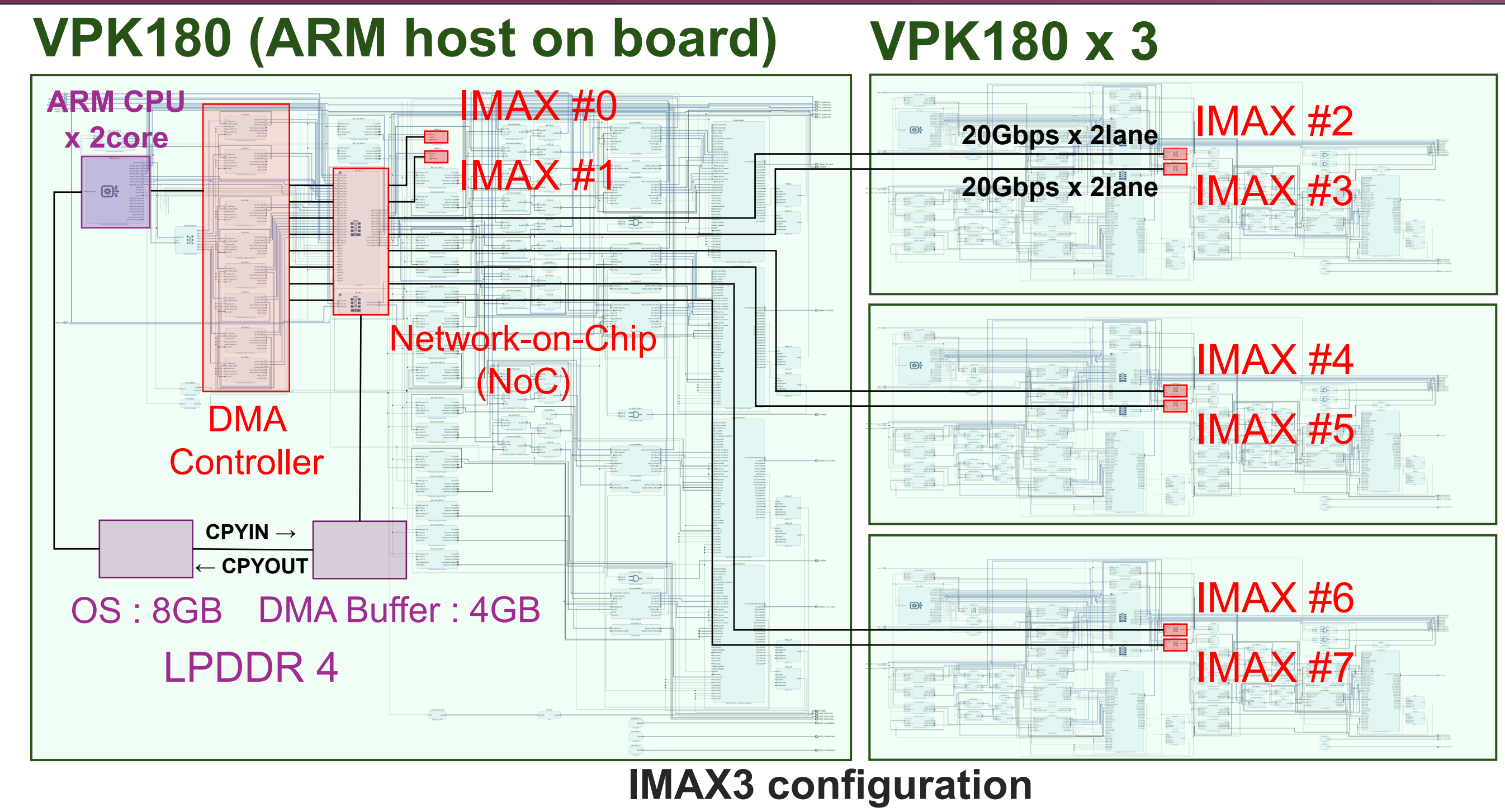
LLM execution on conventional GPUs/CPU drives massive data center power consumption. This demands new, energy-efficient accelerator hardware to achieve sustainable high performance.

IMAX3 Accelerator

- Our proposed solution is **IMAX**[1], a CGRA (Coarse-Grained Reconfigurable Array)-based accelerator designed to offer a superior balance of power efficiency and programmability for LLM inference[2].
- The IMAX3 successfully demonstrated this potential but was **critically limited by its on-board, 2-core ARM host CPU**.

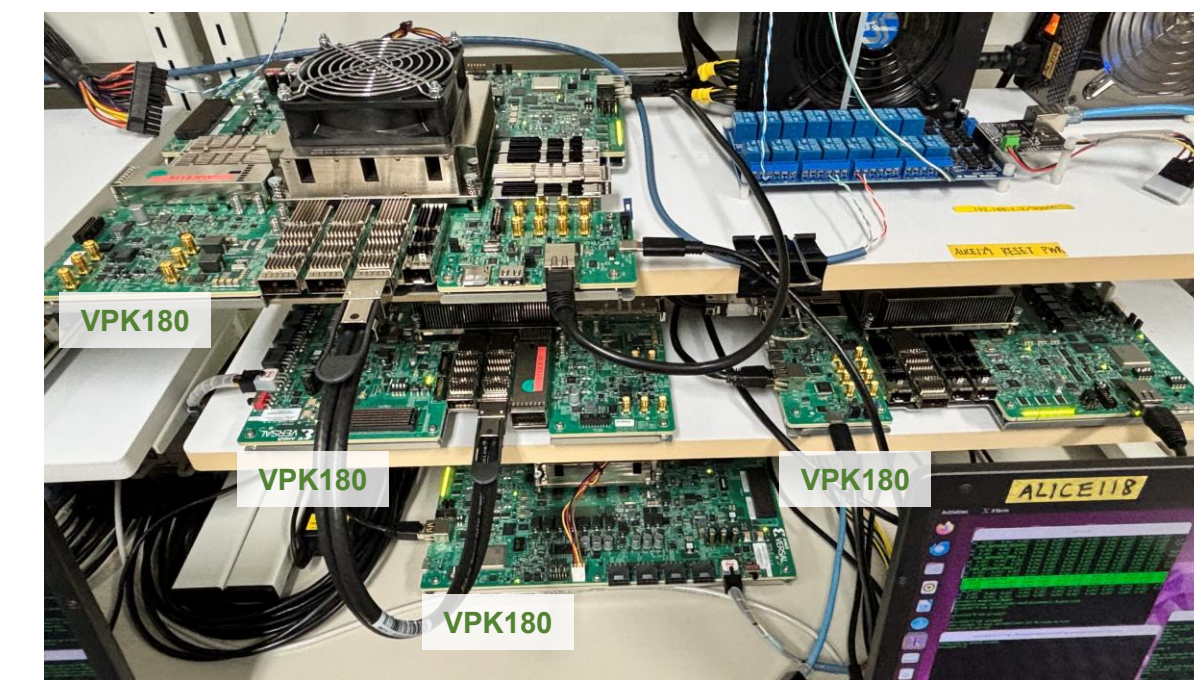
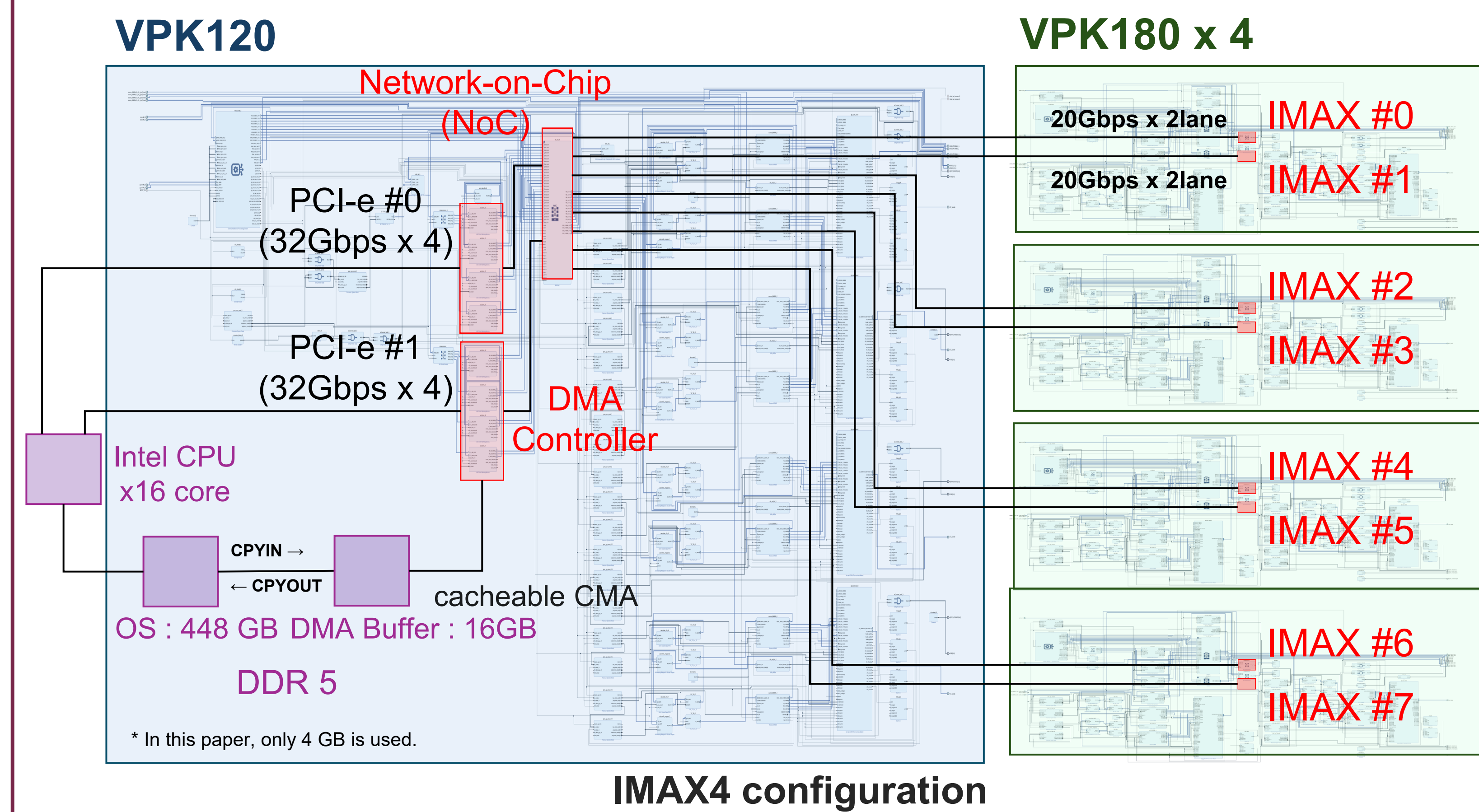
Objective

- To conduct a detailed analysis of IMAX3 using the modern LLaMA3 to quantify the performance limitations imposed by its host bottleneck.
- To implement, and evaluate the IMAX4 prototype, which integrates a server-grade Intel Xeon host and a high-bandwidth PCIe Gen5 interconnect, to demonstrate its effectiveness in overcoming these limitations and validate its potential for server-based LLM acceleration.

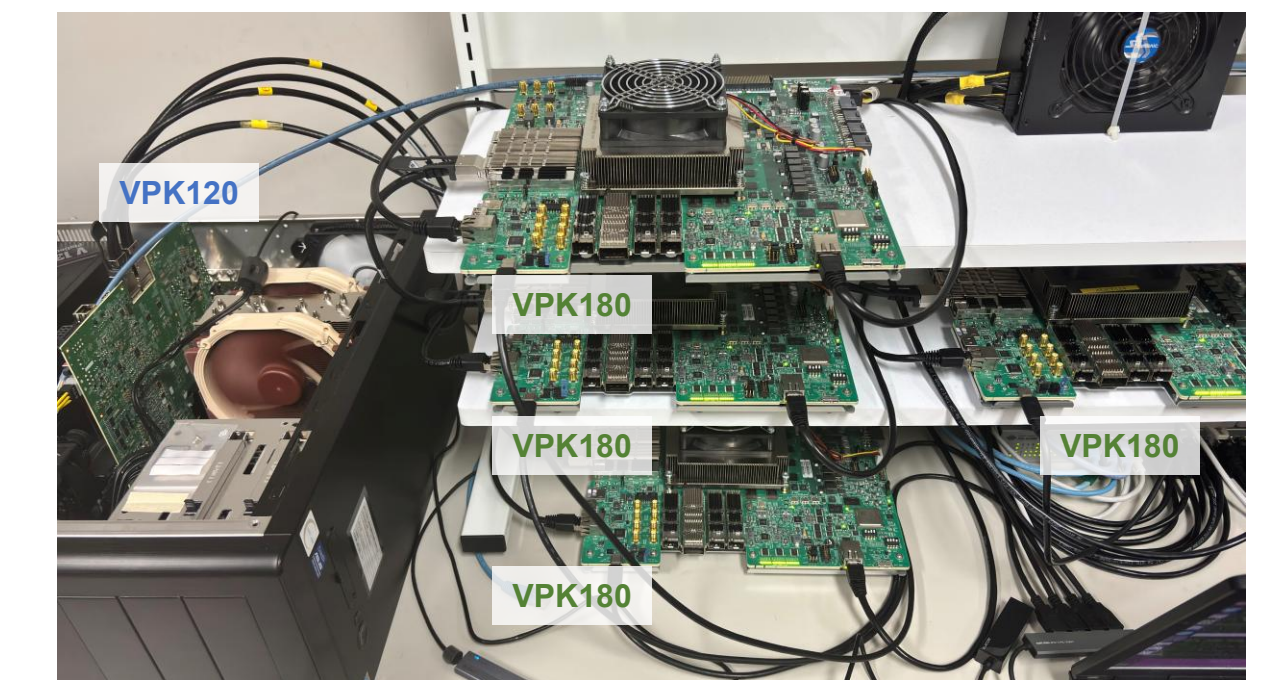


Methods

IMAX4: Methodology for Overcoming Host-Side Bottlenecks



IMAX3 prototype



IMAX4 prototype

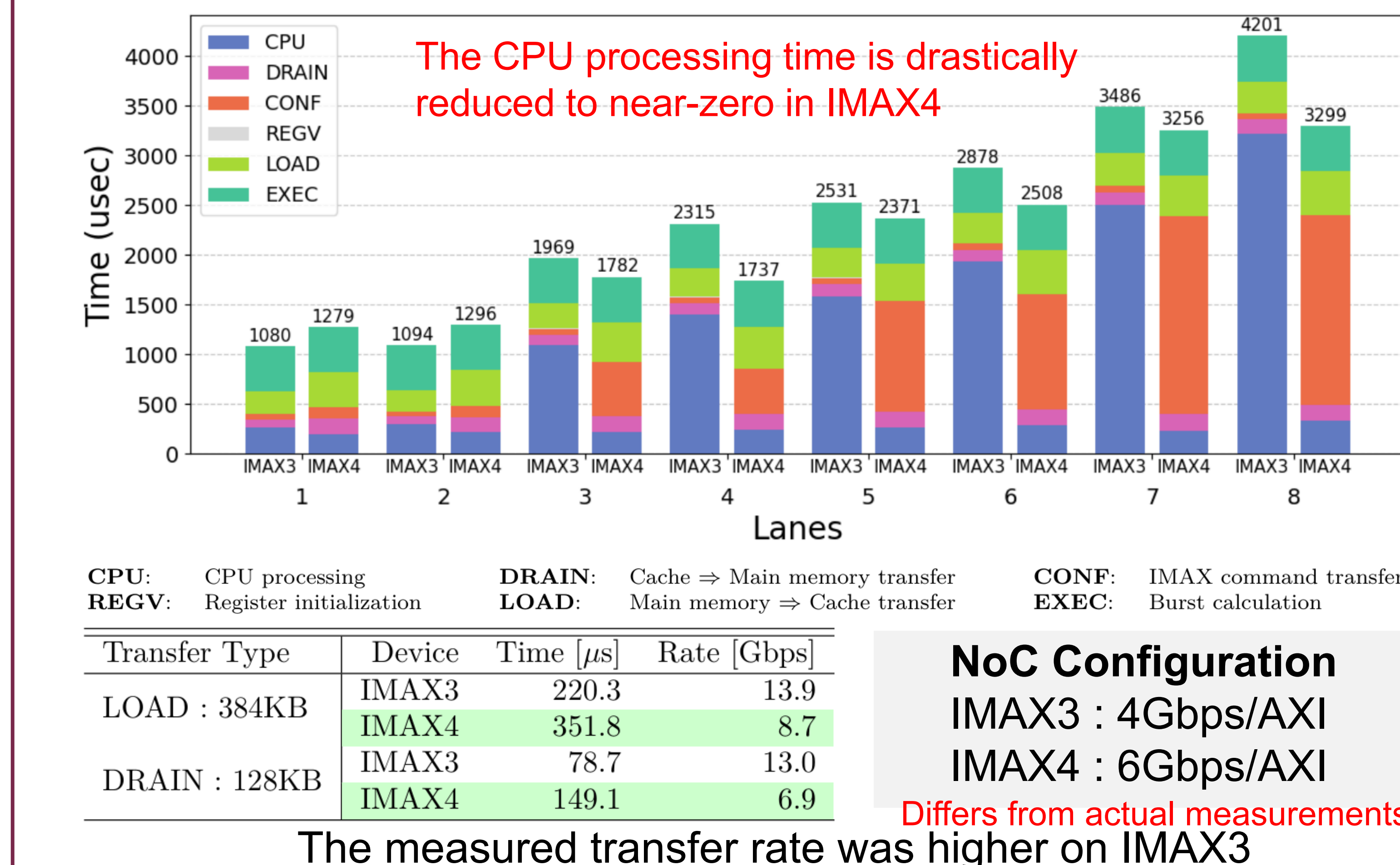
- Architectural Shift:** We transitioned the system from a self-contained SoC model to a **server-oriented**.
- Host Processor Upgrade:** To support the 8-lane IMAX configuration, we upgraded the host to a 16-core processor, **overcoming the severe resource constraints of the IMAX3**

Component	IMAX3	IMAX4
Host CPU	2-core ARM Cortex-A72	16-core Intel Xeon
Memory System	On-board LPDDR4	Host PC's DDR5
Host-FPGA Link	Internal Bus (SoC)	PCIe Gen5 x8

Results

Fundamental Evaluation with Microbenchmarks

We **measured the data transfer and computation performance**. The benchmark involved each lane asynchronously executing a Fused Multiply-Add (FMA) operation ($D = C + A \times B$).



LLM Execution Evaluation with LLaMA3 8B

Model: LLaMA3 8B (Q3_K_S and Q8_0 quantized models)

Conditions: Input: "Hello", Predicted Tokens: 32

Configuration	Device	Lanes	Latency [s]	Processing Time [s]	
				IMAX	CPU
IMAX3	1	88.6	-	64.3	20.4
IMAX4	1	172.0	-	164.1	7.0
IMAX3	2	49.2	-44%	33.3	12.3
IMAX4	2	87.6	-49%	82.6	4.0
IMAX3	3	56.3	-36%	25.9	24.7
IMAX4	3	252.3	+47%	248.4	2.8
IMAX3	4	60.3	-32%	21.6	31.9
IMAX4	4	296.3	+72%	292.5	2.2
IMAX3	5	65.1	-26%	20.2	35.3
IMAX4	5	354.7	+106%	350.9	2.0
IMAX3	6	67.1	-24%	17.5	39.2
IMAX4	6	322.4	+87%	318.2	1.9
IMAX3	7	69.4	-22%	16.9	40.0
IMAX4	7	387.3	+125%	383.7	1.7
IMAX3	8	70.2	-20%	14.4	42.6
IMAX4	8	350.9	+103%	347.3	1.6

IMAX4 Disadvantaged on Lighter Models (Q3_K_S):

Due to **higher data transfer overhead**, IMAX4 exhibited greater latency than IMAX3 on the less computationally intensive model.

IMAX4 Dominates on Heavier Models (Q8_0):

The superior host CPU performance became critical, allowing IMAX4 to **achieve substantially lower latency than IMAX3**.

Conclusion

- The IMAX4 prototype successfully **eliminated the CPU bottleneck** of IMAX3 by decoupling the accelerator from its on-board environment.
- Our analysis revealed that the performance bottleneck has shifted from the host CPU to **the data transfer path**. (Intel Northbridge ?)

References

- T. Akabe, V. Trung Duong LE, and Y. Nakashima, "IMAX: A power-efficient multilevel pipelined cgla and applications," IEEE Access, vol. 13, pp. 31899–31911, 2025..
- Y. Eto and Y. Nakashima, "Implementation and Performance Analysis of LLaMA on a CGLA," in International Conference on Intelligent Systems and Networks (ICISN), 2025.